INTERACTIVE PROJECT REPORT

FACE RECOGNITION ATTENDANCE SYSTEM

Technical documentation, product workflows, recognition pipeline, system controls and quality assurance.



● FACE CAPTURE READY

08	SECTIONS	03	WORKFLOWS

Clickable contents · linked pages · fillable sign-off

PROJECT EDITION / 25 OCTOBER 2026

START HERE

Explore the project

Select a section to jump directly to the content. Use the contents control at the top of each page to return here.

01 **Overview & scope**
Functional summary, verified requirements →

02 **Architecture**
Components and information movement →

03 **User workflows**
Enrollment and attendance flows →

04 **Interface & roles**
Dashboard concepts and permissions →

05 **Data & recognition**
Entity model and vision processing →

06 **Security & operations**
Controls, privacy and troubleshooting →

07 **Testing & deployment**
Acceptance scenarios and go-live sequence →

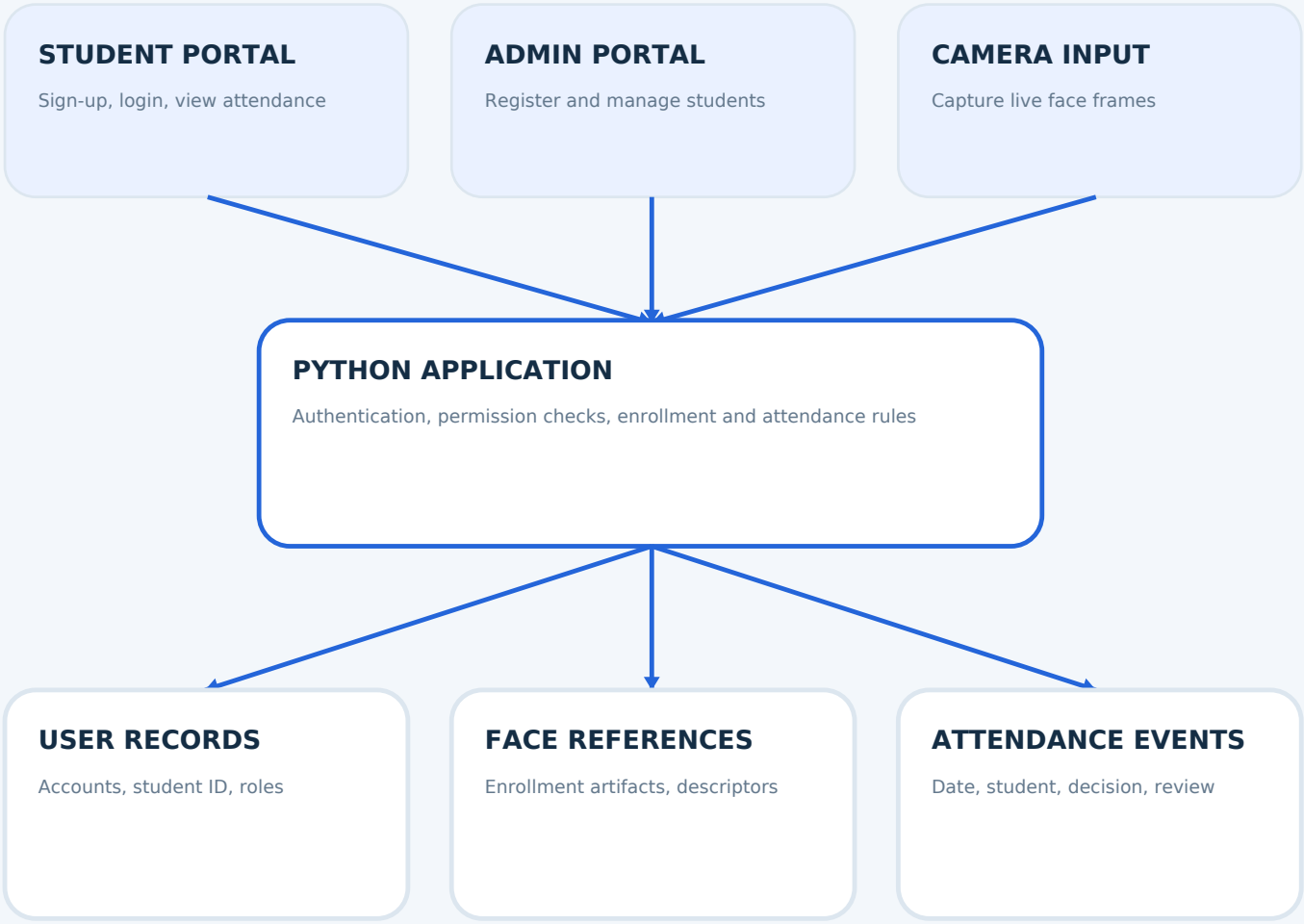
08 **Interactive review**
Fillable project verification sheet →

Scope: conceptual design only until the latest source code is inspected.

01 / ARCHITECTURE

How the system fits together

Reference architecture - component names and exact technologies require source-code confirmation.



ARCHITECTURAL RULE

UI controls are not a security boundary. The application must enforce permissions and validation when data is changed or attendance is recorded.

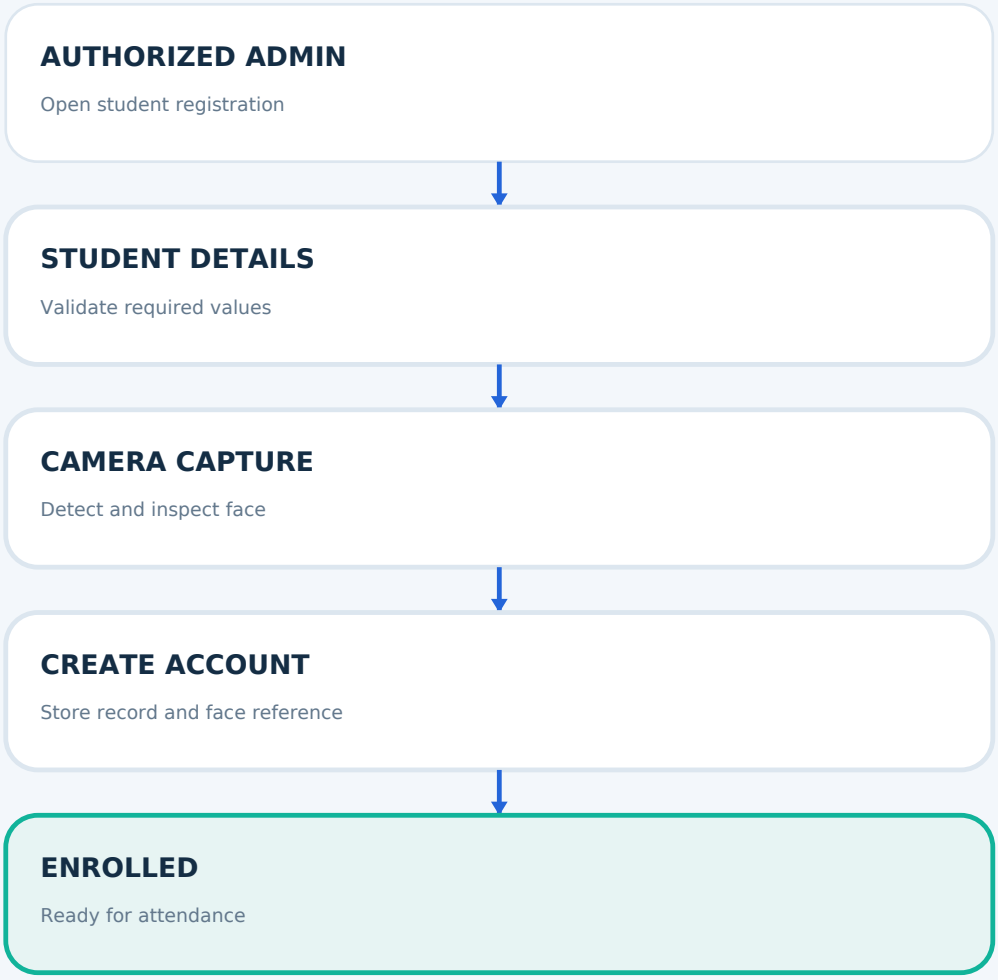
DESIGN RECOMMENDATION

02 / WORKFLOWS

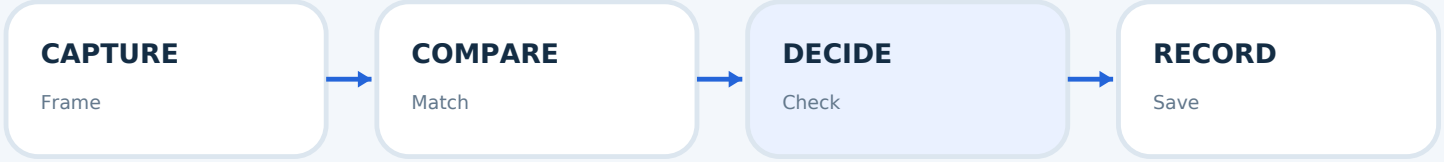
Registration to recognition

Two core journeys, with explicit handling for unsuccessful recognition.

A / ADMIN-ASSISTED ENROLLMENT



B / DAILY ATTENDANCE DECISION

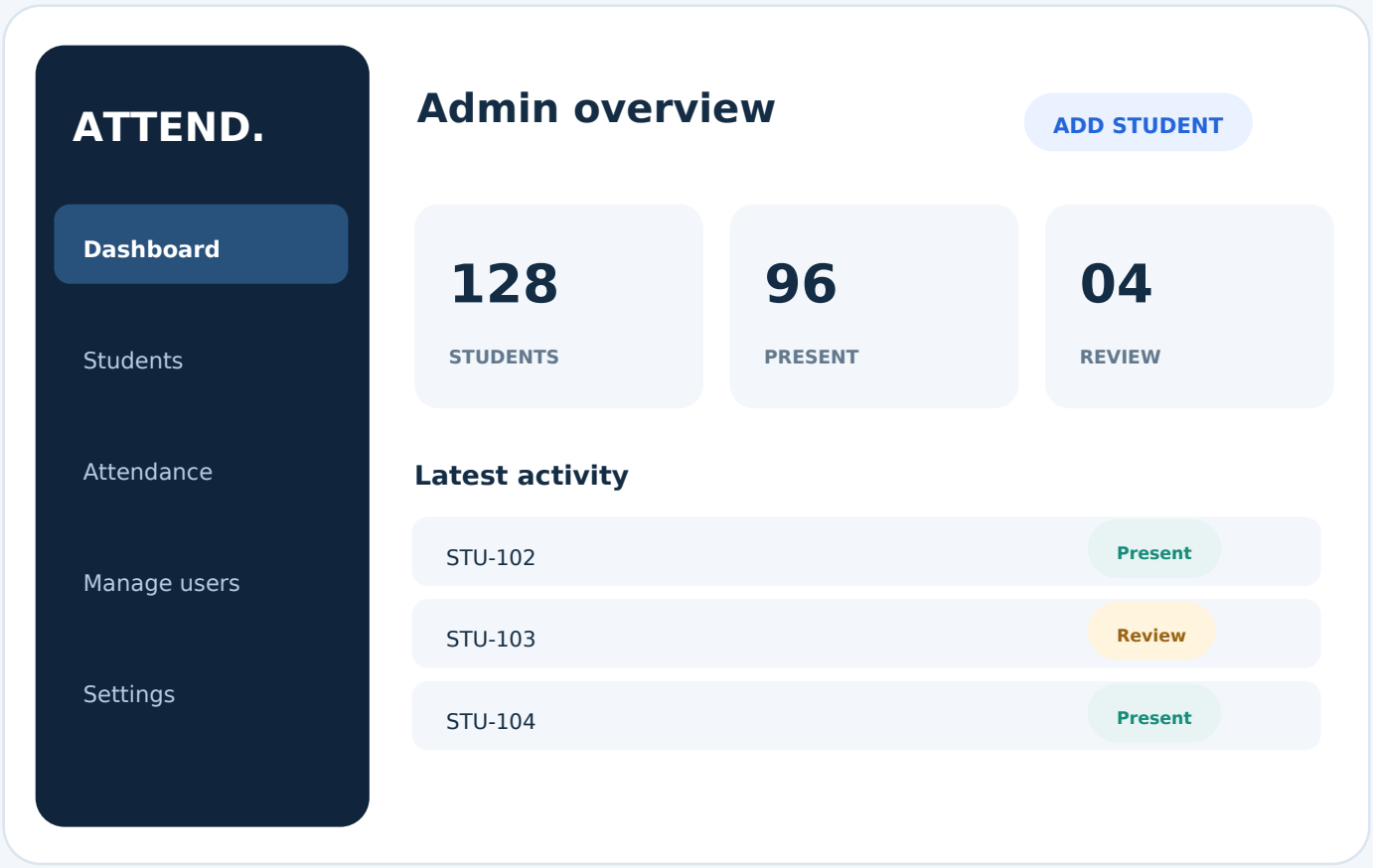


Unknown / weak / duplicate matches should route to review, not auto-mark presence.

03 / INTERFACE

Cleaner dashboard design

Illustrative layouts only - these are not screenshots of the application.



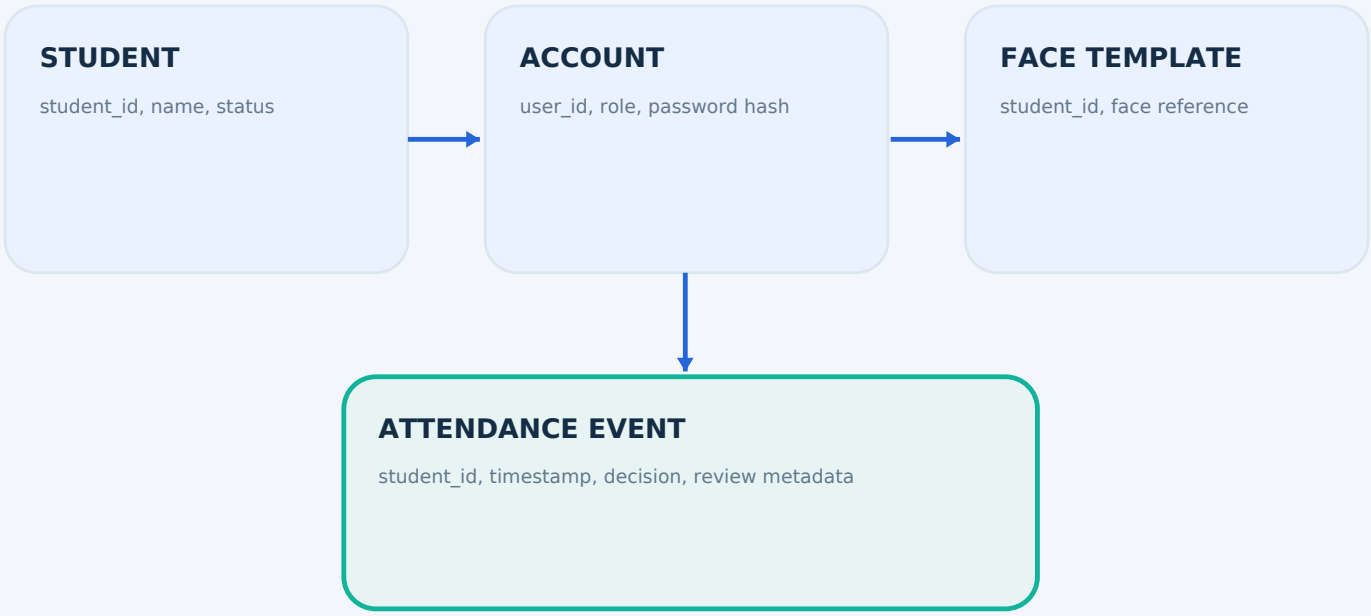
ROLE / CAPABILITY MATRIX

	STUDENT	ADMIN
View own attendance	Yes	**Yes**
Register a student	No	**Yes**
Change user authority	No	**Yes**
Manage users	No	**Yes**

04 / DATA & RECOGNITION

From person to attendance record

Conceptual data entities and biometric pipeline. The current storage engine and model are unverified.



RECOGNITION PIPELINE

01	**Capture**	Acquire camera frame
02	**Detect**	Locate usable face
03	**Compare**	Match enrolled reference
04	**Decide**	Threshold + duplicate rules
05	**Record**	Save or request review

Files noted: face_labels.npz and face_model.yml - do not delete without checking code.

05 / SECURITY & OPERATIONS

Build trust into every step

Important safeguards to consider before relying on a biometric attendance system.

01

Identity & passwords

Store salted password hashes. Force change at first sign-in for default passwords.

02

Server-side permissions

Check authority before registration, user edits and role changes.

03

Biometric consent

Provide notice, consent and a defined retention and deletion policy.

04

Attendance integrity

Reject duplicate events and route uncertain matches for review.

05

Audit trail

Log role changes, account operations and admin attendance overrides.

06

Backup / recovery

Back up database, recognition assets and configuration together.

OPERATIONS QUICK CHECK

Camera unavailable? Check permissions and connection. Button unresponsive? Verify handler, route and access checks.

06 / TESTING & DEPLOYMENT

Acceptance before release

Suggested verification cases based on the requested student and admin behaviors.

01	Student can sign up and access authorized pages
02	Admin registration captures an enrollment face
03	Top-right admin add-student sets Changeme@123 when no password is entered
04	Other registration paths do not inherit this rule inadvertently
05	Student ↔ admin role changes are authorized and recorded
06	Manage Users actions respond and remain correctly aligned
07	Normal page scrolling works; redundant scroll controls are absent
08	Unknown, low-quality and repeated matches do not auto-mark attendance

DEPLOYMENT GATE

Dependencies → Storage → Camera / model → Permissions → End-to-end tests

07 / INTERACTIVE REVIEW

Project sign-off

Fill these fields in a PDF viewer supporting AcroForms. Editable areas are shaded blue.

Reviewer name

Build / version

Reviewed source / commit

Confirmed data storage

Recognition method verified

REVIEW DECISION

- ○ Accepted
- ○ Needs changes
- ◉ Pending verification

REVIEW NOTES